

Scalar E-eigenvalues versus sign-pairs: conjecture 00000002844

earthking11

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The precise counting convention

The source claims that an order- d , n -dimensional symmetric tensor has $((d-1)^n - 1)/(d-2)$ E-eigenvalues, without identifying opposite scalar values or imposing a dimension lower bound. We use standard normalized scalar E-eigenvalues. For a real tensor complexified to $\mathbb{C}^n$, these satisfy

$$\sum_i x_i^2 = 1, \quad x \neq 0, \quad (Tx^{d-1})_i = \lambda x_i.$$

The normalization uses the complex-bilinear extension of the Euclidean quadratic form, not a Hermitian norm. The primary definition calls the scalar λ itself an E-eigenvalue. For odd order, (λ, x) and $(-\lambda, -x)$ are both normalized eigenpairs. See Sodomaco, Definition 1.1 [1].

Theorem 1. *The one-dimensional order-three symmetric tensor $T_{111} = 1$ has exactly two distinct scalar E-eigenvalues, 1 and -1 . The displayed formula equals 1, so it does not count these E-eigenvalues as stated.*

Proof. There is one tensor coefficient, so all permutations of its three indices leave it unchanged: it is symmetric. The actual tensor contraction is $Tx^2 = x^2$. The normalized eigenvalue equations reduce to

$$x^2 = 1, \quad x \neq 0, \quad x^2 = \lambda x.$$

The factorization $(x-1)(x+1) = 0$ over $\mathbb{C}$ implies $x = 1$ or $x = -1$. Substitution then gives $\lambda = 1$ or $\lambda = -1$, respectively. Conversely each of these pairs satisfies every equation. Thus the actual scalar-value set is exactly $\{1, -1\}$, with cardinality 2. At $d = 3, n = 1$, the asserted formula is $((3-1)^1 - 1)/(3-2) = 1$, not 2. $\square$

What this does and does not refute

This is a disproof of the source's literal scalar-E-eigenvalue count. It is not a counterexample to a formula counting values modulo the identification $\lambda \sim -\lambda$: in this example there is exactly one sign-pair, agreeing with the displayed number. The standard odd-order theorem indeed counts sign-pairs, while the E-characteristic polynomial has twice that pair-count degree [1]. Confusing these two quantities is precisely the issue. The Lean proof establishes the actual scalar-value set and its cardinality; it does not assert that it formalizes a resultant construction or its algebraic degree. If the intended conjecture is instead sign-pair counting, that altered convention must be stated, and this counterexample does not apply to it. No such identification appears in the source.

Formal verification

Lean defines the actual tensor on `Fin 1`, its symmetry, double-sum contraction, quadratic unit and nonzero conditions, eigenpair equations, and scalar-value set with vector witnesses. It proves both directions of the exact set characterization and then computes its cardinality. The project builds on Lean 4.33.1 with pinned Mathlib, without unfinished proofs, native evaluation, extra axioms, or long brute force. Its audited dependencies are propositional extensionality, classical choice, and quotient soundness.

References

- [1] Luca Sodomaco, *The Product of the Eigenvalues of a Symmetric Tensor*, arXiv:1802.10173, 2018. Definition 1.1, p. 1; Theorem 1.3 and the subsequent degree discussion, p. 3. <https://arxiv.org/abs/1802.10173>.